Lecture 3

Registers: the state of a part

Objectives. *Lecture 2 read a whole register, and only at the end of a circuit. This lecture reads part of one, and asks:* what is the state of part of a register? *The answer, the reduced state (Theorem 3.9), is the one matrix that gives every measurement of the part the statistics the whole register gives; it has a formula in any basis. For the Bell state the answer is* $\frac{1}{2}\mathbb{1}$*, which no ket describes.*

By the end of this lecture you will be able to:

- Build a register from its parts as a tensor product, and decide whether a state of two qubits is a product.
- Compute the probabilities and the states left by a reading of part of a register, and by a measurement in any basis, drawn as a circuit.
- Prove that a global phase changes nothing in any procedure, while a relative phase can be seen.
- Define the reduced state by its property, prove it unique, derive its formula, and compute it for the Bell state.

3.1 Tensor structure and entanglement

Let a register AB of $m+n$ qubits have the part A on its top m wires and the part B on the n wires below. For columns $\psi \in \mathbb{C}^{2^m}$ and $\varphi \in \mathbb{C}^{2^n}$, the Kronecker product of Definition 1.2 is the column $\psi \otimes \varphi \in \mathbb{C}^{2^{m+n}}$ with the entries

$$(\psi \otimes \varphi)_{xx'} = \psi_x \varphi_{x'},$$

one for each string xx' of AB; it is linear in each factor, and is also written $|\psi\rangle\,|\varphi\rangle$. By the interchange law and (2.3),

$$\begin{aligned} \langle \psi \otimes \varphi | \psi' \otimes \varphi' \rangle &= (\psi^\dagger \otimes \varphi^\dagger)(\psi' \otimes \varphi') \\ &= \langle \psi | \psi' \rangle \, \langle \varphi | \varphi' \rangle \,, \end{aligned} \tag{3.1}$$

so a product of states is a state, and $(U \otimes V)(\psi \otimes \varphi) = U\psi \otimes V\varphi$: a gate on one part acts on its own factor. The columns $|x\rangle\,|x'\rangle = |xx'\rangle$ are a basis, by (1.3), so every state of AB is $\sum_{x,x'} c_{xx'}\,|x\rangle\,|x'\rangle$. In the language of *Linear Algebra* (L12), $\mathbb{C}^{2^{m+n}}$ is the tensor product $\mathbb{C}^{2^m} \otimes \mathbb{C}^{2^n}$, realised by Kronecker products. Not every state of AB is a product.

Definition 3.1 (Product, entangled)**.** A state of AB is a *product* if it is $\psi \otimes \varphi$ for a state ψ of A and a state φ of B; otherwise it is *entangled*.

Proposition 3.2 (A test for two qubits)**.** *A state* $\sum_{j,k} c_{jk}\,|jk\rangle$ *of two qubits is a product if and only if* $c_{00}c_{11} = c_{01}c_{10}$.

Proof. If it is $\psi \otimes \varphi$, then $c_{jk} = \psi_j \varphi_k$, and both sides are $\psi_0\psi_1\varphi_0\varphi_1$. Conversely, the condition says that the 2×2 matrix (c_{jk}), which is not zero, has determinant zero, so it has rank one: its rows are $\psi_0 u^{\mathsf{T}}$ and $\psi_1 u^{\mathsf{T}}$ for one non-zero column u. Then $c_{jk} = \psi_j u_k$, and the state is $\psi \otimes u$. By (3.1), $\|\psi\|\,\|u\| = 1$, so it is the product of the states $\psi/\|\psi\|$ and $\|\psi\|\,u$. ∎

Example 3.3 (QQ: a product, and an entangled state)**.** The state

$$|+\rangle\,|+\rangle = \tfrac{1}{2}\big(|00\rangle + |01\rangle + |10\rangle + |11\rangle\big)$$

has $c_{00}c_{11} = \frac{1}{4} = c_{01}c_{10}$: it is a product, although every string appears in it. The Bell state

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

has $c_{00}c_{11} = \frac{1}{2}$ and $c_{01}c_{10} = 0$: it is entangled.

Griffiths and Schroeter (p. 199) first describe entanglement by dependence: the state of one particle "depends on" the state of the other. Their definition, which follows, is this course's: not a product. A superposition of every string can still be a product.

3.2 Measuring part of a register

Definition 2.1 read a whole register as Griffiths' measurement (3.43) of an observable with 2^n distinct eigenvalues, each of multiplicity one. A reading of the part A has degenerate outcomes: the outcome k is shared by every string kx'. Griffiths' rule does not say what state such a reading leaves; its collapse "to the corresponding eigenstate" (p. 102) presumes there is only one. The rule that does say is Lüders'.

Definition 3.4 (Projective measurement, Lüders' rule)**.** A *projective measurement* of a register is a family of matrices $\Pi_1, \ldots, \Pi_K$ with $\Pi_k^\dagger = \Pi_k$, $\Pi_j\Pi_k = \delta_{jk}\Pi_k$ and $\sum_k \Pi_k = \mathbb{1}$. On the state ψ it gives the outcome k with probability

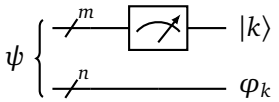
$$p_k = \langle\psi|\,\Pi_k\,|\psi\rangle = \|\Pi_k\psi\|^2\,,$$

and, if $p_k > 0$, it leaves the state $\Pi_k\psi/\sqrt{p_k}$ (*Lüders' rule*).

The probabilities add up to $\langle\psi|\psi\rangle = 1$. Reading the whole register is the case $\Pi_x = |x\rangle\langle x|$: it gives x with probability $|\psi_x|^2$, as in Lecture 2, and leaves $(\psi_x/|\psi_x|)\,|x\rangle$, which is $|x\rangle$ up to a factor of modulus one; Theorem 3.7 shows that such a factor changes nothing. Reading A is the measurement $\Pi_k = |k\rangle\langle k| \otimes \mathbb{1}$, for $k \in \{0,1\}^m$. On $\psi = \sum_{k,x'} c_{kx'}\,|k\rangle\,|x'\rangle$ it gives

$$\begin{aligned} p_k &= \textstyle\sum_{x'} |c_{kx'}|^2\,, \\ \Pi_k\psi/\sqrt{p_k} &= |k\rangle \otimes \varphi_k, \\ \varphi_k &= p_k^{-1/2} \textstyle\sum_{x'} c_{kx'}\,|x'\rangle\,, \end{aligned} \tag{3.2}$$

drawn as a meter on the wires of A alone, which carry $|k\rangle$ on:



Reading B afterwards gives x' with probability $|c_{kx'}|^2/p_k$, so the two readings give the pair (k, x') with probability $|c_{kx'}|^2$, as one reading of the whole register does.

Example 3.5 (QQ: reading one qubit)**.** Read the top qubit of $|+\rangle|+\rangle$. The outcome 0 has probability $\frac{1}{2}$, and its eigenspace, spanned by $|00\rangle$ and $|01\rangle$, has dimension two; Lüders' rule leaves $|0\rangle|+\rangle$, and the qubit not read is as it was. Read the top qubit of Φ^+: it gives 0 or 1 with probability $\frac{1}{2}$ each, and leaves $|00\rangle$ or $|11\rangle$. The qubit not read now holds $|k\rangle$, and a reading of it gives k with certainty.

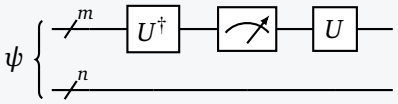
The word "collapse" names this update. It is not a dynamics: it is the rule by which the predictions for later steps are conditioned on the outcome.

Other bases. For a gate U on A, the basis $\{U|k\rangle\}$ of A (every orthonormal basis is one) gives the projective measurement

$$\Pi_k^U = U|k\rangle\langle k|U^\dagger \otimes \mathbb{1},$$

the measurement of A *in the basis* $\{U|k\rangle\}$.

Proposition 3.6 (A measurement in a basis, as a circuit)**.** *In the circuit*



(i) $U^\dagger$ *and the reading of* A *give* k *with the probability* $\left\|\Pi_k^U\psi\right\|^2$*;*

(ii) the reading leaves A *in* $|k\rangle$*, and* U *applied after it leaves the Lüders state* $\Pi_k^U\psi/\left\|\Pi_k^U\psi\right\|$.

Proof. By the interchange law, $\Pi_k^U = (U\otimes\mathbb{1})(|k\rangle\langle k|\otimes\mathbb{1})(U^\dagger\otimes\mathbb{1})$. Since $U\otimes\mathbb{1}$ is unitary, $\left\|\Pi_k^U\psi\right\|$ is the norm of $(|k\rangle\langle k|\otimes\mathbb{1})(U^\dagger\otimes\mathbb{1})\psi$, whose square is the probability of reading k after $U^\dagger$: this is (i). By (3.2) the reading leaves A in $|k\rangle$, and $U\otimes\mathbb{1}$ turns the state it leaves into $\Pi_k^U\psi$ divided by the same norm: this is (ii). ∎

So $U^\dagger$ and a reading give the probabilities of the measurement in the basis $\{U|k\rangle\}$, but leave $|k\rangle$, not $U|k\rangle$. For one qubit and $U = H$ the basis is $|\pm\rangle$, the eigenvectors of X. Nielsen and Chuang (p. 22) say that this measurement leaves $|+\rangle$ or $|-\rangle$, which is the Lüders state; their recipe of Ex. 4.33, a change of basis and then a reading, gives the circuit of H and a reading, which leaves $|0\rangle$ or $|1\rangle$. In this course, "measure in the basis $\{U|k\rangle\}$" means the projective measurement, and its circuit is $U^\dagger$, a reading, and U.

3.3 Phases

Lecture 2 promised that the sign $(-1)^{f(0)}$ left by Deutsch's circuit (Example 2.6) changes nothing, whatever follows. A *procedure* is a sequence of steps, each a gate, the joining of a fresh register in a given state, or a projective measurement; each step may be chosen according to the outcomes before it.

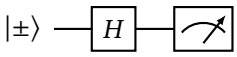
Theorem 3.7 (Global and relative phases)**.** *Let* ψ *be a state and* α *real.*

(i) For every procedure, the runs on ψ *and on* $e^{i\alpha}\psi$ *give every sequence of outcomes with the same probability, and so does a run in which a gate* U *of the procedure is replaced by* $e^{i\alpha}U$.

(ii) $|+\rangle$ *and* $|-\rangle$*, read, give* 0 *and* 1 *with probability* $\frac{1}{2}$ *each;* H *and a reading tell them apart with certainty.*

Proof. (i) By induction on the number of steps, with the claim: every sequence h of outcomes so far has the same probability in the two runs, and if it is positive, the states it leaves are ψ_h and $c_h\psi_h$ with $|c_h| = 1$. At the start, c is $e^{i\alpha}$, or 1 when a gate is replaced. The step after h is the same in both runs, since it depends on h only. A gate sends $c_h\psi_h$ to $c_hU\psi_h$, and $e^{i\alpha}U$ sends it to $e^{i\alpha}c_hU\psi_h$; a joined register gives $c_h(\psi_h \otimes \varphi)$; a projective measurement gives k with the probability $\|\Pi_k c_h\psi_h\|^2 = \|\Pi_k\psi_h\|^2$, and leaves c_h times the state it leaves from ψ_h. (ii) The amplitudes of $|\pm\rangle$ are $\pm 1/\sqrt{2}$, and $H|+\rangle = |0\rangle$, $H|-\rangle = |1\rangle$ by (2.2), since $H^2 = \mathbb{1}$. ∎

The predictions depend on a state only through its *ray*, the set of its multiples $e^{i\alpha}\psi$, and the sign in Example 2.6 may be dropped. A *relative* phase, the sign of one amplitude of $|-\rangle$, is seen by (ii):



By (i), a gate may be replaced by $e^{i\alpha}$ times itself, as $U \simeq V$ (equal up to a global phase) allows; inside a controlled gate it may not, since there the phase becomes relative (Lecture 6).

3.4 Density operators recalled, the reduced state

Density operators, recalled (Griffiths–Schroeter §12.3). The density operator of a state ψ is $|\psi\rangle\langle\psi|$, their (12.14). If the state ψ_i is prepared with probability p_i, the *mixture* has the density operator

$$\rho = \textstyle\sum_i p_i |\psi_i\rangle\langle\psi_i| ,$$

their (12.28). Since $\langle\psi|\Pi|\psi\rangle = \mathrm{Tr}(\Pi|\psi\rangle\langle\psi|)$, averaging over i the rules for gates and for projective measurements gives them in terms of ρ alone: a gate U takes ρ to $U\rho U^\dagger$, and a projective measurement gives

$$\begin{aligned} p_k &= \mathrm{Tr}(\rho\Pi_k), \\ \rho &\mapsto \Pi_k\rho\Pi_k/p_k. \end{aligned} \tag{3.3}$$

The update is Lüders' rule averaged by Bayes' rule: given k, with $p_k > 0$, each ψ_i with $\Pi_k\psi_i \neq 0$ has the probability $p_i\langle\psi_i|\Pi_k|\psi_i\rangle/p_k$ and has become $\Pi_k\psi_i/\|\Pi_k\psi_i\|$, while the others have probability 0 and contribute nothing to $\Pi_k\rho\Pi_k$. The expectation of an observable M is $\mathrm{Tr}(\rho M)$, their (12.33). And $e^{i\alpha}\psi$ has the same density operator as ψ: the density operator is the ray of Theorem 3.7, made into a matrix.

Definition 3.8 (Density operator)**.** A *density operator* on $\mathbb{C}^d$ is a self-adjoint $d \times d$ matrix ρ with $\langle v|\rho|v\rangle \geq 0$ for every v and $\mathrm{Tr}\,\rho = 1$. It is *pure* if $\rho = |\psi\rangle\langle\psi|$ for a state ψ, and *mixed* otherwise.

Every mixture is one, since $\mathrm{Tr}\,\rho = \sum_i p_i = 1$ and $\langle v|\rho|v\rangle = \sum_i p_i |\langle\psi_i|v\rangle|^2$. Conversely, by the spectral theorem (*Linear Algebra*, L10), $\rho = \sum_j \lambda_j |e_j\rangle\langle e_j|$ with the e_j orthonormal, and the $\lambda_j = \langle e_j|\rho|e_j\rangle \geq 0$ add up to $\mathrm{Tr}\,\rho = 1$: every density operator is a mixture. A pure $|\psi\rangle\langle\psi|$ has the eigenvalue 1, on ψ, and 0 on the vectors orthogonal to ψ.

The state of a part. Let AB be in the state ρ. A projective measurement $\{\Pi_k\}$ of A alone is the measurement $\{\Pi_k \otimes \mathbb{1}\}$ of AB, with the probabilities $\mathrm{Tr}(\rho(\Pi_k \otimes \mathbb{1}))$. A state of A should predict these, and the expectation of every observable of A, as the state of the whole does.

Theorem 3.9 (The reduced state)**.** *Let ρ be a density operator on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$.*

(i) There is exactly one $d_A \times d_A$ matrix ρ_A with

$$\mathrm{Tr}(\rho_A M) = \mathrm{Tr}\big(\rho(M \otimes \mathbb{1})\big) \tag{3.4}$$

for every $d_A \times d_A$ matrix M. It is the reduced state *of A, written* $\operatorname{Tr}_B \rho$.

(ii) For every orthonormal basis $f_1, \dots, f_{d_B}$ of $\mathbb{C}^{d_B}$,

$$\rho_A = \textstyle\sum_b \iota_b^\dagger \rho \iota_b, \tag{3.5}$$

where $\iota_b = \mathbb{1} \otimes |f_b\rangle$ is the Kronecker product of the $d_A \times d_A$ identity and the column f_b. In the computational basis, $(\rho_A)_{aa'} = \sum_b \langle ab| \rho |a'b\rangle$.

(iii) ρ_A is a density operator.

Proof. (i) If σ and σ' both satisfy (3.4), then $D = \sigma - \sigma'$ has $\operatorname{Tr}(DM) = 0$ for every M. For $M = |a'\rangle\langle a|$ this says $\langle a| D |a'\rangle = 0$, so $D = 0$.

(ii) By (2.3), $\iota_b^\dagger = \mathbb{1} \otimes \langle f_b|$, and by the interchange law $\iota_b M = M \otimes |f_b\rangle = (M \otimes \mathbb{1})\iota_b$. Let σ be the right side of (3.5). Since $\operatorname{Tr}(XY) = \operatorname{Tr}(YX)$ for a $p \times q$ matrix X and a $q \times p$ matrix Y,

$$\begin{aligned} \operatorname{Tr}(\sigma M) &= \textstyle\sum_b \operatorname{Tr}\big(\rho \iota_b M \iota_b^\dagger\big) \\ &= \textstyle\sum_b \operatorname{Tr}\big(\rho (M \otimes \mathbb{1}) \iota_b \iota_b^\dagger\big) \\ &= \operatorname{Tr}\big(\rho (M \otimes \mathbb{1})\big), \end{aligned}$$

because $\sum_b \iota_b \iota_b^\dagger = \mathbb{1} \otimes \sum_b |f_b\rangle\langle f_b| = \mathbb{1}$. So σ satisfies (3.4), which proves existence, and $\sigma = \rho_A$ by (i), whatever the basis. For $f_b = |b\rangle$, $\iota_b |a'\rangle = |a'b\rangle$ gives the entries.

(iii) $M = \mathbb{1}$ gives $\operatorname{Tr}\rho_A = \operatorname{Tr}\rho = 1$. Each term $\iota_b^\dagger \rho \iota_b$ is self-adjoint, and $\langle v| \rho_A |v\rangle = \sum_b \langle v \otimes f_b| \rho |v \otimes f_b\rangle \geq 0$, since $\iota_b v = v \otimes f_b$. ∎

Both sides of (3.4) are linear in M, and every matrix is $M_1 + iM_2$ with $M_1 = (M + M^\dagger)/2$ and $M_2 = (M - M^\dagger)/2i$ self-adjoint; so asking (3.4) of every observable is asking it of every matrix. One observable is not enough: $\frac{1}{2}\mathbb{1}$ and $|+\rangle\langle +|$ give the reading in Z the same probabilities, $\frac{1}{2}$ and $\frac{1}{2}$, while $\operatorname{Tr}(\frac{1}{2}X) = 0$ and $\langle +| X |+\rangle = 1$. Nielsen and Chuang define Tr_B by its values on products, their (2.178), and show in Box 2.6 that it is the only map with the property (3.4); here the property is the definition, and the formula is derived. The state $\rho_B = \operatorname{Tr}_A \rho$ of B is defined in the same way, with $\mathbb{1} \otimes M$.

Corollary 3.10 (A part of a pure state)**.** *Let $f_1, \dots, f_{d_B}$ be an orthonormal basis of $\mathbb{C}^{d_B}$, and let χ be a state of AB. Then $\chi = \sum_b \chi_b \otimes f_b$ with $\chi_b = \iota_b^\dagger \chi$, and*

$$\operatorname{Tr}_B |\chi\rangle\langle\chi| = \textstyle\sum_b |\chi_b\rangle\langle\chi_b| .$$

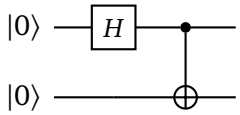
In particular, the part A of a product $\psi \otimes \varphi$ is in the pure state $|\psi\rangle\langle\psi|$.

Proof. Since $\sum_b \iota_b \iota_b^\dagger = \mathbb{1}$, $\chi = \sum_b \iota_b \chi_b = \sum_b \chi_b \otimes f_b$, and the terms of (3.5) for $\rho = |\chi\rangle\langle\chi|$ are $\iota_b^\dagger |\chi\rangle\langle\chi| \iota_b = |\chi_b\rangle\langle\chi_b|$. For the product, take a basis with $f_1 = \varphi$: by the interchange law $\iota_b^\dagger(\psi \otimes \varphi)$ is $\psi \langle f_b|\varphi\rangle$, which is ψ for $b = 1$ and 0 otherwise. ∎

Measuring B in the basis $\{f_b\}$ is the projective measurement $\Pi_b = \iota_b \iota_b^\dagger = \mathbb{1} \otimes |f_b\rangle\langle f_b|$, and $\Pi_b \chi = \chi_b \otimes f_b$: it gives b with probability $\|\chi_b\|^2$ and, if that is positive, leaves A in $\chi_b/\|\chi_b\|$. So $\operatorname{Tr}_B |\chi\rangle\langle\chi|$ is the mixture of these states with these probabilities, and it is the same for every basis of B.

3.5 The Bell state

Example 3.11 (QQ: the Bell state and its half)**.** A Hadamard, then a CNOT, makes Φ^+ (Nielsen and Chuang, §1.3.6):



$$= \left|\Phi^+\right\rangle \tag{3.6}$$

an equality of vectors: $(H \otimes \mathbb{1}) |00\rangle = |+\rangle |0\rangle$, which is $(|00\rangle + |10\rangle)/\sqrt{2}$, and the CNOT turns $|10\rangle$ into $|11\rangle$. In the computational basis of B, $\Phi^+ = (|0\rangle/\sqrt{2}) \otimes |0\rangle + (|1\rangle/\sqrt{2}) \otimes |1\rangle$, so by Corollary 3.10

$$\begin{aligned} \mathrm{Tr}_B \left|\Phi^+\right\rangle \left\langle\Phi^+\right| &= \tfrac{1}{2} |0\rangle \langle 0| + \tfrac{1}{2} |1\rangle \langle 1| \\ &= \tfrac{1}{2}\mathbb{1}, \end{aligned}$$

and $\mathrm{Tr}_A |\Phi^+\rangle \langle\Phi^+| = \frac{1}{2}\mathbb{1}$ in the same way.

The half has no ket. If a state ψ of one qubit predicted every measurement of A, so that, for $\rho = |\Phi^+\rangle \langle\Phi^+|$, $\mathrm{Tr}(\rho(M \otimes \mathbb{1})) = \langle\psi| M |\psi\rangle = \mathrm{Tr}(|\psi\rangle \langle\psi| M)$ for every M, then by the uniqueness in Theorem 3.9, $\frac{1}{2}\mathbb{1} = |\psi\rangle \langle\psi|$; but $|\psi\rangle \langle\psi|$ has the eigenvalue 1, and $\frac{1}{2}\mathbb{1}$ has only $\frac{1}{2}$. In every basis $\{U|k\rangle\}$ the half gives each outcome with probability $\mathrm{Tr}(\frac{1}{2}U|k\rangle \langle k|U^\dagger) = \frac{1}{2}$, as its reading in Example 3.5 did. In Z this matches $|+\rangle$; in X it does not, so $|+\rangle$ is not the half's state. Griffiths and Schroeter assert this matrix, their (12.40), for one particle of an entangled pair, which "does not occupy a pure state"; here both are derived. The pair is in a pure state, yet its half has no ket: the state of a part is $\mathrm{Tr}_B \rho$.

Summary.

- *Handed on:* registers as tensor products, and entangled as not a product (Definition 3.1 and Proposition 3.2); Lüders' rule for reading a part, and the measurement in the basis $\{U|k\rangle\}$ as $U^\dagger$, a reading and U (Definition 3.4 and Proposition 3.6).
- *Phases:* a global phase changes nothing in any procedure; a relative one can be seen (Theorem 3.7).
- *The main result:* the state of a part is $\mathrm{Tr}_B \rho$, the one matrix that predicts every measurement of the part, with a formula in every basis (Theorem 3.9 and Corollary 3.10).
- *QQ:* the Bell state is pure, and its halves are in $\frac{1}{2}\mathbb{1}$, which no ket describes (Example 3.11).
- *Next:* Lecture 4, what can be done with an unknown state.

References

Griffiths and Schroeter, pp. 102–103, 199 and 455–459 (§12.3). Nielsen and Chuang, pp. 16–17, 22, 25–26, 71–74, 87–88, 93–96, 99–107 (§2.4; Box 2.6, p. 107) and 187–188 (Ex. 4.33). Watrous (2025), pp. 43–53 (little-endian), 85–86 and 279–283. Kitaev, Shen and Vyalyi, Def. 10.2, p. 96. G. Lüders, Ann. Phys. (Leipzig) **8** (1951), 322–328.